Windows Client Onboarding Manual

Squid Transparent HTTPS Proxy, HSTS Prevention & Root CA Installation

Document Version: 1.1 (Updated for HSTS Fixes)	**Target Platform:** Windows 10 / Windows 11 / Windows Server
Proxy Server: QNAP NAS (192.168.1.90)	**Router Gateway:** ASUS Router (192.168.0.1)

1. Architecture & HSTS Prevention

This environment uses **Transparent Network Interception**. Windows clients do **NOT** require manual proxy configurations in System Settings or Internet Options.

- **Router Redirection:** The ASUS Router detects TCP traffic on ports 80 (HTTP) and 443 (HTTPS) from registered client IPs and routes it to the Squid proxy at `192.168.1.90`.
- **SSL Bumping:** Squid inspects TLS traffic and dynamically generates site-specific certificates signed by the internal Root CA (`squid.local`).
- **Preventing HSTS Hard-Blocks (e.g., `docs.google.com`):** Popular domains like Google, GitHub, and Twitter enforce *HTTP Strict Transport Security (HSTS)*. If a browser encounters an untrusted Root CA on an HSTS domain, it **hard-blocks access** without allowing a bypass option. Installing the Root CA into the **Local Machine Trusted Root Store** ensures Chrome, Edge, and system services recognize `squid.local` as a valid Root Authority, eliminating HSTS errors.

> **Key Windows Behavior:** Unlike Linux, **Microsoft Edge, Google Chrome, Brave, Opera, and Vivaldi on Windows all natively read from the Windows System Certificate Store (`Cert:\LocalMachine\Root`)**. Installing the Root CA into `LocalMachine\Root` instantly fixes HSTS errors for all Chromium-based browsers on Windows.

2. Step 1: Router Registration (Administrator Action)

Before a Windows client can be transparently proxied, its IP address must be registered in the proxy host list.

1.1 Assign Fixed IP Address

Assign a static IP or DHCP Reservation on the ASUS Router (**LAN** → **DHCP Server** → **Manually Assigned IP**).

1.2 Add Host to Proxy List

On your Linux management workstation, edit `squid/router/proxy-hosts.conf`:

```
# Format: IP_ADDRESS HOSTNAME
192.168.8.30   vm-server1
192.168.8.40   windows-pc1    # Add Windows client IP and hostname here
```

1.3 Deploy Router Rules

Execute the deployment script to apply policy routing rules to the router:

```
./squid-mgmt.sh router-deploy
```

3. Step 2: Install Root CA Certificate on Windows Client

To prevent `NET::ERR_CERT_AUTHORITY_INVALID` and HSTS hard-blocks, install the Root CA into the System-Wide Local Machine store.

2.1 Obtain Certificate File

Copy `certs/squid-ca.pem` from your repository to the Windows PC and rename it to `squid-ca.crt`.

Method A: Automated PowerShell Installation (Recommended)

Open **PowerShell as Administrator** on the Windows machine and execute:

```
Import-Certificate -FilePath "C:\path\to\squid-ca.crt" -CertStoreLocation "Cert:\LocalMachine\Root"
```

Verification: PowerShell output will display `Cert:\LocalMachine\Root` and Subject `CN=squid.local, OU=Proxy, O=Home LAN`.

Method B: Graphical Interface (GUI) Installation

1. Press `Win + R`, type **`certlm.msc`** (Local Computer Certificate Manager), and press **Enter**.
2. In the left pane, expand **Trusted Root Certification Authorities** → **Certificates**.
3. Right-click **Certificates** → **All Tasks** → **Import...**
4. Click **Next**, browse to `squid-ca.crt`, and select **Place all certificates in the following store: Trusted Root Certification Authorities**.
5. Click **Next** → **Finish**. Confirm the security prompt if shown.

Method C: Group Policy (Active Directory Environments)

For domain-joined Windows computers, deploy the certificate via Active Directory GPO:

```
Computer Configuration → Policies → Windows Settings → Security Settings →
Public Key Policies → Trusted Root Certification Authorities → Import
```

4. Step 3: Browser-Specific Setup & Clearing HSTS Caches

4.1 Microsoft Edge & Google Chrome

Chrome and Edge automatically trust the certificate once imported into `Cert:\LocalMachine\Root`. If you visited an HSTS website (like `docs.google.com`) *before* installing the CA certificate:

1. Close and re-open all browser windows.
2. If the error persists due to cached HSTS state, navigate to `chrome://net-internals/#hsts` (or `edge://net-internals/#hsts`).
3. Under **Delete domain security policies**, enter `docs.google.com` and click **Delete**.

4.2 Mozilla Firefox on Windows

By default, Firefox maintains an isolated certificate store. To make Firefox trust the Windows System Certificate Store:

1. Open Firefox and type `about:config` in the address bar.
2. Search for **`security.enterprise_roots.enabled`** and double-click to set it to **`true`**.
3. Restart Firefox. It will now trust `squid.local` automatically.

5. Step 4: Verification & Testing

5.1 Web Browser Test

1. Open Chrome/Edge and navigate to `https://docs.google.com` or `https://google.com`.
2. Click the **Padlock Icon** → **Connection is secure** → **Certificate is valid**.
3. Verify Issuer: `CN=squid.local, OU=Proxy, O=Home LAN`.

5.2 Terminal Test (PowerShell)

```
Invoke-WebRequest -Uri "https://docs.google.com" -UseBasicParsing
```

5.3 Real-Time Access Log Verification

On your Linux management workstation, verify intercepted requests from the Windows client IP:

```
./squid-mgmt.sh catlogs
```

6. Troubleshooting Quick Reference

Symptom / Issue	Cause	Resolution
HSTS error on `docs.google.com`	CA cert missing from Local Machine store or cached HSTS block.	Ensure cert is in `Cert:\LocalMachine\Root`. Delete domain policy in `chrome://net-internals/#hsts`.
`SEC_ERROR_UNKNOWN_ISSUER` in Firefox	Firefox system store integration disabled.	Set `security.enterprise_roots.enabled` = `true` in `about:config`.
Traffic bypasses Squid proxy	Browser using HTTP/3 (QUIC over UDP 443).	The router REJECTs UDP 443 to force TCP 443 fallback. Disable QUIC in `chrome://flags/#enable-quic` if needed.